

Yuhan Zhao

Ph.D. Candidate · Cyberspace Security · Expected Jun 2027

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Research Profile

I study the implementation security of post-quantum and conventional public-key cryptography, with a focus on ML-DSA and ECDSA. My work combines side-channel attack development, leakage analysis, and countermeasure design and evaluation on embedded devices.

Education

Nanjing University of Science and Technology

Sep 2022 – Present

Ph.D. Candidate, Cyberspace Security; expected graduation: Jun 2027.

Advisors: Prof. Yongbin Zhou and Prof. Wei Cheng

Qilu University of Technology (Shandong Academy of Sciences)

Sep 2019 – Jun 2022

M.Eng., Computer Application Technology

Jiangsu Police Institute

Sep 2014 – Jun 2018

B.Eng., Network Security and Law Enforcement

Publications and Manuscripts

C Conference · J Journal · S Submitted/preprint · T Thesis · * Corresponding author

- [C.1] Yuhan Zhao, Dalin He, Wei Cheng, Yuejun Liu, Jingdian Ming, and Yongbin Zhou. *When Module Lattice Leaks: Horizontal Fusion Attacks on ML-DSA Implementation*. **ACM CCS 2026**. Accepted; to appear.
- [C.2] Yuhan Zhao, Zehua Qiao, Wei Cheng, Yuejun Liu, and Yongbin Zhou. *Deep Learning-based Public Template Attack against ML-DSA on ARM Microcontrollers*. **IEEE ISCAS 2026**, pp. 420–424.
- [C.3] Yuhan Zhao, Wei Cheng, Zehua Qiao, Yuejun Liu, and Yongbin Zhou. *Rejection Matters: Efficient Non-Profiling Side-Channel Attack on ML-DSA via Exploiting Public Templates*. **DATE 2026**, pp. 1–7. IEEE.
- [C.4] Zelong Zhang, Wei Cheng, Yongbin Zhou, Zehua Qiao, Yuhan Zhao, and Jian Weng. *Attacking High-order Masked Cryptosystem via Deep Learning-based Side-Channel Analysis*. **IEEE TrustCom 2024**, pp. 699–706.
- [C.5] Yan Zhou, Huiling Shi, Yuhan Zhao, Wei Gao, and Wei Zhang. *Encrypted Network Traffic Identification Based on 2D-CNN Model*. **IEEE APNOMS 2021**, pp. 238–241.
- [C.6] Wei Gao, Meihong Yang, Wei Zhang, Yuhan Zhao, Yan Zhou, and Kai Zhang. *FFDNet-Based Channel Estimation for Beamspace mmWave Massive MIMO Systems*. **IEEE ISPA 2021**, pp. 1167–1170.
- [C.7] Meihong Yang, Wei Zhang, Yuhan Zhao, and Lizhuang Tan. *PANGU: A Cloud-Edge Collaborative Resource Management Platform Centered on Supercomputing*. **ACM CoNEXT 2020**, pp. 552–553.
- [C.8] Yuhan Zhao, Wei Zhang, Meihong Yang, and Huiling Shi. *Network Resource Scheduling for Cloud/Edge Data Centers*. **IEEE IPCCC 2020**, pp. 1–4.
- [J.1] Jingshan Pan, Zhiqiang Wei, Yuhan Zhao*, Yan Zhou, Xunyu Lin, Wei Zhang, and Chang Tang. *Enhanced FCN for Farmland Extraction from Remote Sensing Image*. *Multimedia Tools and Applications*, 81(26):38123–38150, 2022.
- [J.2] Hanyu Yang, Xutao Li, Wenhao Qiang, Yuhan Zhao, Wei Zhang, and Chang Tang. *A Network Traffic Forecasting Method Based on SA Optimized ARIMA-BP Neural Network*. *Computer Networks*, 193:108102, 2021.
- [J.3] Yuhan Zhao, Wei Zhang, Longquan Zhou, and Wenpeng Cao. *A Survey on Caching in Mobile Edge Computing*. *Wireless Communications and Mobile Computing*, 2021:5565648, 2021.
- [S.1] Zehua Qiao, Yuejun Liu, Yongbin Zhou, Yuhan Zhao, and Shuyi Chen. *Single Trace Is All It Takes: Efficient Side-Channel Attack on Dilithium*. *Cryptology ePrint Archive*, Report 2024/512, 2024.
- [S.2] Yuhan Zhao, Wei Cheng, Yuejun Liu, Yongbin Zhou, and Wei Wei. *Manuscript under review*.
- [T.1] *Research on Edge Caching Strategy Based on Content Popularity Prediction*. Master's thesis, 2022.

Research and Project Experience

Key Technologies for In-Depth Information Leakage Analysis and Reliable Protection of Cryptographic Chips

Key Project Member

National Key R&D Program of China; No. 2022YFB3103800

Jun 2023 – Present

- **Task 1: toolkit delivery.** Responsible for testing, performance optimization and packaging of methods into a toolkit.
- **Task 5: testbed integration.** Integrated, tested and tuned methods; achieved **over 50× speedup** for TVLA leakage detection. Integrated ISCAS, DATE and CCS methods as operational testbed deliverables.
- **ECDSA smart cards: 75% fewer profiling traces** and **3× the key-recovery success rate** versus the comparison method; **successful cross-device attack**.
- **ISCAS 2026 [C.2]:** Deep-learning public-template attack on ML-DSA on ARM microcontrollers; **50% fewer signatures** than SOTA under the **same acquisition settings**.
- **DATE 2026 [C.3]:** Non-profiling ML-DSA attack using public rejection-sampling information and public templates; **90% fewer signatures** and **3× speedup** versus the comparison method.
- **CCS 2026 [C.1]:** ML-DSA horizontal fusion attack; **at least 70% fewer signatures** unprotected and **at least 50% fewer with masking**, relative to the comparison methods.

5G-Based Applications for Representative Scenarios

Project Member

Shandong Provincial Key R&D Program

Sep 2019 – May 2021

Shandong Academy of Sciences; National Supercomputing Center in Jinan

Contributed to dynamic edge-node scheduling and request handling; optimized task allocation for backend resource efficiency; contributed to heterogeneous-device integration and unified management.

Undergraduate Research Mentoring

Non-Profiling Attack Methods for the Post-Quantum Signature Algorithm ML-DSA

Lead research mentor for an undergraduate project selected as a **national-level project**.

Research and Technical Skills

- **Programming and platforms:** Python and C (proficient); C++ and Java (working knowledge); Linux, macOS and Windows.
- **Side-channel security:** Attack analysis; countermeasure design, optimization, deployment and evaluation for ML-DSA and ECDSA; ARM Cortex-M and FPGA security evaluation.
- **Technical writing:** Grant proposals, academic papers and project completion reports in English and Chinese; ~~LaTeX~~ and Markdown.

Awards and Honors

Second Prize, 9th National College Cryptography and Mathematics Challenge	2024
National Graduate Scholarship, awarded in two consecutive years	2020–2021
Second/Third Prize, MathorCup Graduate Mathematical Modeling Challenge	2020–2021
Third Prize, Shandong Higher Education Science and Technology Award	2019
Second Prize, 1st National Police College Teaching Skills Competition	2017

Languages and Certifications

College English Test Band 6 (CET-6): 616	Dec 2015
IELTS 7.5 (L 9.0, S 6.0, R 8.5, W 6.0); GRE 335	2019/2021
CATTI English Translation Level II; Translation and Interpretation Level III	2017–2020
NCRE Level 4 Network Engineer (Excellent)	2017
French: DELF B1, 79.5/100	2025